Array transfer

Given data in the form of an array *Arr* of size *N*. You have to transfer a subsequence of size *M* to the client.

The time taken to transfer a subsequence of size *M* is the product of the *first* and *last* element of the subsequence.

As you do not want your client to wait, you want to minimize the time taken to transfer the subsequence of size *M*.

What is the minimum time the client has to wait for a subsequence of size *M?*

Function description

Complete the function *solve()*. This function takes the following *3* parameters and returns the required answer:

- *N:* Represents an integer denoting the size of the given array *Arr*
- *M:* Represents an integer denoting the size of the subsequence
- *Arr:* Represents an integer array of size *N* representing the

```
int solve(vector<int> &v,int m){
    int n=v.size();
    vector<int> preMin(n);
    preMin[0]=v[0];
    vector<int> sufMin(n);
    sufMin[n-1]=v[n-1];
    for(int i=1;i<n;i++){
        preMin[i]=min(preMin[i-1],v[i]);
    }
    for(int i=n-2;i>=0;i--){
        sufMin[i]=min(sufMin[i+1],v[i]);
    }
    int ans =INT_MAX;
    int i=0;
    int j=m-1;
    while(i<=n-m){
        ans = min(ans,preMin[i]*sufMin[j]);
        i++;
        j++;
    }
    return ans;
}
```

Range game

You are at the range. You have N targets labeled 1 to N from left to right.

Hit the target by performing the following operation:

- In each round starting from the first element and moving from left to right, hit every alternate target.
- Repeat the above step until only one target is left.

Note: You will get i points when you hit a target labeled *i*.

Calculate the score you will obtain by continuously hitting targets until there is only one target remaining.

Function description

Complete the *Solve()* function. This function takes the following argument and returns the total number of points you will get by hitting targets until only one target is left:

- N: Represents the total number of targets

```
 int Josephus(int n,int k){
     int ans=0;
     for(int i=1;i<=n;i++){
         ans = (ans+k)%i;
     }
     return ans+1;
}
```

Train destinations

You are given data on different types of trains and their departure and destination stations in the given database.

Task

Write an SQL query to find the names and types of trains that have destinations in Pennsylvania or Chicago. Sort the final output by *train_name*.

Table description

Input format

Table: train_details

Name	Type	Description
train_name	string	Represents the name of a train
train_type	int	Represents the ID of the type of train
train_from	int	Represents the ID of the departing station of the train
train_to	int	Represents the ID of the destination station of the train

Table: train_type_details

```sql
SELECT train_name, train_type
FROM trains
WHERE destination IN ('Pennsylvania', 'Chicago')
ORDER BY train_name;
```

In an operating system, consider the following scenario.

Scenario:

There are six resources. 'p' number of processes are competing for these resources. Each process may require three resources.

What is the maximum value of 'p' such that no deadlock occurs at all?

- 4
- 3
- 2
- 1

In an operating system, which of the following ways for implementing the Page Table can be used?

Options:

1. Dedicated registers
2. In the main memory
3. Associative registers (Cache)
4. Secondary memory

- 1, 2, and 4
- 1, 2, and 3
- 1, 3, and 4
- All of these

Reset Answer

You are given the following information:

A is the brother of E.

M is the daughter of A and also the sister of Q.

T is the mother of Q.

V is the father of T.

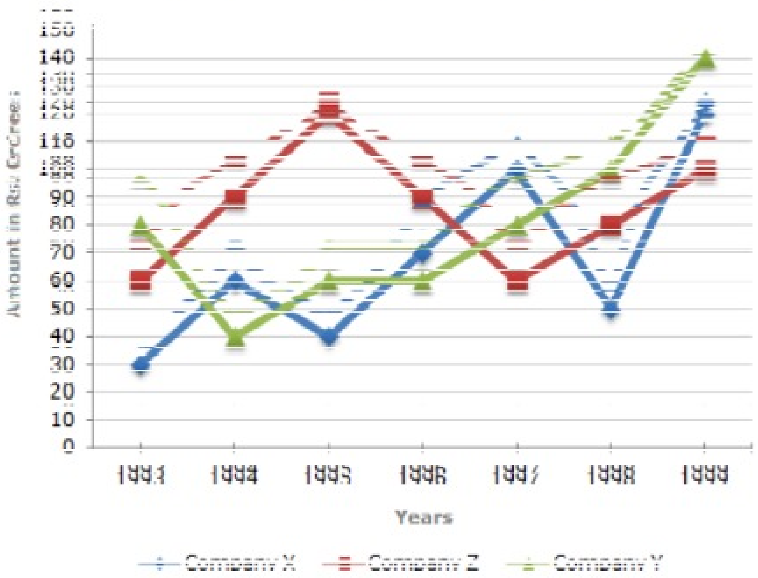
P, who is the son of H is the brother of T.

D has only one son and is married to K, who is the mother of A.

Now based on the given information, how is P related to Q?

- Uncle
- Father
- Brother-in-law
- Son

Study the following line graph and answer the questions.



In how many of the given years shown in the line graph, were the exports from Company Z below the average annual exports over the given years?

In how many of the given years shown in the line graph, were the exports from Company Z below the average annual exports over the given years?

2

3

A, B, C, D, E, F, G, and H are sitting in a round circle. A is sitting at a position second to the right of E who is the neighbor of C and G. There are 2 people in between A and G and 3 people in between A and F. If A and D are not neighbors and H is between B and D then who is to the right of F?

- A
- G
- D
- C

Reset Answer